25900567 **A**

Reg.No : .....

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION October 2025

Third SEMESTER

Value Addition Course (VAC) - MG3VACPHY200 - SCIENCE AND SOCIETY

(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 Minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Objective Type Questions

Answer any **25** questions

Each question carries **2** marks

1. Approximate value of earth's circumference? [K] / [CO1]

   a). 400 km b). 4000 km

   c). 40000 km d). Not calculated yet

2. The first human in space was [K] / [CO2]

   a). Yuri Gagarin b). Neil Armstrong

   c). Valentina Tereshkova d). Buzz Aldrin

3. Which of the following is not an outcome of scientific thinking in daily life ? [U] / [CO4]

   a). Environmental Awareness b). Emotional Reasoning

c). Creativity and Innovation d). Problem-solving Skills

4. Name of our galaxy is : [K] / [CO1]

a). Andromeda b). Whirlpool

c). Milkyway d). Triangulum

5. Identify 'Optical Instrument' given in below options [U] / [CO2]

a). Microscope b). Telescope

c). Camera d). All of the above

6. The term “scientific temper” was popularized in India by: [U] / [CO4]

a). Dr. A.P.J. Abdul Kalam b). Jawaharlal Nehru

c). Vikram Sarabhai d). C.V. Raman

7. The discovery that the universe is expanding shows that scientific theories can change when: [U] / [CO1]

a). Scientist makes random guesses

b). Ancient beliefs are followed strictly

c). Personal opinions of scientists change

d). New Evidence is found

8. What is the main goal of vaccination? [K] / [CO2]

a). To prevent specific diseases before they happen

b). To directly kill harmful bacteria

c). To reduce pain and fever

d). To replace antibiotics

9. Assertion (A): Radiation always causes instant mutations visible on the body. [U] / [CO5]
Reason (R): Most mutations occur at the DNA level and may not be immediately visible.

a). Both A and R are true, and R is the correct explanation of A

b). Both A and R are true, but R is not the correct explanation of A

c). A is true, but R is false

d). A is false, but R is true

10. The Flat Earth theory declined mainly due to [U] / [CO1]

a). Philosophical debates

b). Scientific observations and measurements

c). Religious approval

d). Mythological stories

11. A large collection of data is called a: [K] / [CO2]

a). Dataset

b). Document

c). Module

d). Network

12. Match the myth with the correct fact [U] / [CO5]

Column A	Column B
a) Moon phases caused by Earth's shadow	1) moon phases are a result of the changing angles of sunlight reflecting off the Moon as it orbits Earth
b) Dark side of Moon is always dark	2) we only see one side of the Moon from Earth, but both sides experience day and night
c) Stars are tiny dots nearby	3) Stars are giant objects and they appear tiny to us as they are located very far away
d) Space is empty	4) Space contains gas, dust, radiation and other particles

a). a–2, b–1, c–4, d–3

b). a–1, b–2, c–3, d–4

c). a–4, b–3, c–1, d–2

d). a–4, b–2, c–1, d–3

13. Why is scientific inquiry said to be open-ended? [U] / [CO1]

a). A) It avoids new data

b). B) It concludes easily

c). C) It constantly seeks deeper understanding

d). D) It rejects uncertainty

14. What makes quantum computers special compared to regular computers? [U] / [CO2]

a). They are smaller in size

b). They use electricity more efficiently

c). They follow the strange rules of quantum physics

d). They have larger screens

15. Assertion (A): Humans and chimpanzees share a common ancestor. Reason (R): DNA evidence shows ~98% similarity in sequences. [K] / [CO5]

a). Both A and R are true, and R is the correct explanation of A

b). Both A and R are true, but R is not the correct explanation of A

c). A is true, but R is false

d). A is false, but R is true

16. A theory that cannot be tested or disproven is considered: [U] / [CO1]

a). Scientific

b). Verified

c). Unscientific

d). Proven true

17. Graphene, which is a nanomaterial, is an allotrope of which element? [K] / [CO2]

a). Silicon

b). Boron

c). Nitrogen

d). Carbon

18. Which statement about solar eclipses is a fact? [U] / [CO5]

a). Eclipses can cause natural disasters like earthquakes and tsunamis.

b). Animals behave strangely because an eclipse emits special rays that affect them

c). Looking at the Sun directly during an eclipse can cause permanent eye damage

d). Solar eclipses are rare and happen only once every few centuries.

19. An example of a scientific hypothesis provided is: [U] / [CO1]

a). F=ma.

b). The Earth revolves around the Sun.

c). Increasing the amount of sunlight will speed up the growth of sunflower plants

d). Matter cannot be created or destroyed.

20. Which material is most commonly used for making semiconductors? [K] / [CO2]

a). Copper

b). Silicon

c). Iron

d). Gold

21. A faulty thermometer that reads 2°C higher than actual will always produce: [K] / [CO6]

a). Random error

b). Systematic error

c). Both errors

d). No error

22. What was a significant consequence for Galileo due to his strong support for Copernicus's heliocentric model? [U] / [CO1]

a). He received widespread immediate acceptance and accolades.

b). He was appointed as a chief advisor to the Church.

c). He faced conflict with the Church and was eventually tried for heresy.

d). He became the first person to travel to the Moon.

23. What is the main purpose of a nuclear power plant? [U] / [CO2]

a). To produce electricity

b). To produce coal

c). To produce natural gas

d). To produce solar energy

24. The uncertainty in measuring a pendulum's time period is reduced by [K] / [CO6]

a). Using a stopwatch with larger least count

b). Measuring the time for multiple oscillations

c). Taking a single measurement

d). Measuring at random times

25. Which term refers to evidence supporting a theory? [U] / [CO1]

a). Corroboration

b). Contradiction

c). Hypothesis

d). Law

26. Which of the following is a negative environmental impact of science and technology? [U] / [CO3]

a). Electronic waste (E-waste) generation

b). Resource Depletion

c). Carbon Emissions from wide use of Technology

d). All of the above

27. Which of the following best describes a spurious measurement? [U] / [CO6]

a). A measurement taken with the smallest possible error.

b). A measurement that deviates significantly from the rest of the data set.

c). A measurement taken multiple times for consistency.

d). A measurement rounded off to the nearest integer

28. Which early modern scientist's observations of the phases of Venus supported the heliocentric model, challenging the ancient geocentric worldview? [U] / [CO1]

a). Galileo Galilei

b). Tycho Brahe

c). Johannes Kepler

d). Nicolaus Copernicus

29. Which of the following is an ethical issue in genetic modification? [K] / [CO3]

a). Low cost of technology

b). High speed of results

c). Creation of genetically modified organisms (GMOs) without public consent

d). Use of traditional farming methods

30. Assertion (A): Random errors shift all measured values in one direction. Reason (R): Averaging reduces random errors [U] / [CO6]

a). Both A and R are true, and R is the correct explanation of A

b). Both A and R are true, but R is not the correct explanation of A

c). A is true, but R is false

d). A is false, but R is true

[2x25 = 50]

25900567 **B**

Reg.No : .....

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION October 2025

Third SEMESTER

Value Added Course (VAC) - MG3VACPHY200 - SCIENCE AND SOCIETY

(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 Minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Objective Type Questions

Answer any **25** questions

Each question carries **2** marks

1. Which material is most commonly used for making semiconductors? [K] / [CO2]

   a). Copper

   b). Silicon

   c). Iron

   d). Gold

2. A faulty thermometer that reads 2°C higher than actual will always produce: [K] / [CO6]

   a). Random error

   b). Systematic error

   c). Both errors

d). No error

3. Assertion (A): Radiation always causes instant mutations visible on the body. [U] / [CO5]
Reason (R): Most mutations occur at the DNA level and may not be immediately visible.

a). Both A and R are true, and R is the correct explanation of A

b). Both A and R are true, but R is not the correct explanation of A

c). A is true, but R is false

d). A is false, but R is true

4. Approximate value of earth's circumference? [K] / [CO1]

a). 400 km b). 4000 km

c). 40000 km d). Not calculated yet

5. Why is scientific inquiry said to be open-ended? [U] / [CO1]

a). A) It avoids new data

b). B) It concludes easily

c). C) It constantly seeks deeper understanding

d). D) It rejects uncertainty

6. Name of our galaxy is : [K] / [CO1]

a). Andromeda b). Whirlpool

c). Milkyway d). Triangulum

7. What makes quantum computers special compared to regular computers? [U] / [CO2]

a). They are smaller in size

b). They use electricity more efficiently

c). They follow the strange rules of quantum physics

d). They have larger screens

8. Identify 'Optical Instrument' given in below options [U] / [CO2]

a). Microscope b). Telescope

c). Camera d). All of the above

9. Which of the following is an ethical issue in genetic modification? [K] / [CO3]

a). Low cost of technology

b). High speed of results

c). Creation of genetically modified organisms (GMOs) without public consent

d). Use of traditional farming methods

10. [K] / [CO2]

A large collection of data is called a:

a). Dataset

b). Document

c). Module

d). Network

11. Which of the following is not an outcome of scientific thinking in daily life ? [U] / [CO4]

a). Environmental Awareness b). Emotional Reasoning

c). Creativity and Innovation d). Problem-solving Skills

12. Match the myth with the correct fact [U] / [CO5]

Column A Column B

a) Moon phases caused by Earth's shadow 1) moon phases are a result of the changing angles of sunlight reflecting off the Moon as it orbits Earth

b) Dark side of Moon is always dark 2) we only see one side of the Moon from Earth, but both sides experience day and night

c) Stars are tiny dots nearby 3) Stars are giant objects and they appear tiny to us as they are located very far away

d) Space is empty 4) Space contains gas, dust, radiation and other particles

a). a–2, b–1, c–4, d–3

b). a–1, b–2, c–3, d–4

c). a–4, b–3, c–1, d–2

d). a–4, b–2, c–1, d–3

13. The Flat Earth theory declined mainly due to [U] / [CO1]

a). Philosophical debates

b). Scientific observations and measurements

c). Religious approval

d). Mythological stories

14. An example of a scientific hypothesis provided is: [U] / [CO1]

a). F=ma.

b). The Earth revolves around the Sun.

c). Increasing the amount of sunlight will speed up the growth of sunflower plants

d). Matter cannot be created or destroyed.

15. What is the main goal of vaccination? [K] / [CO2]

a). To prevent specific diseases before they happen

b). To directly kill harmful bacteria

c). To reduce pain and fever

d). To replace antibiotics

16. A theory that cannot be tested or disproven is considered: [U] / [CO1]

a). Scientific

b). Verified

c). Unscientific

d). Proven true

17. Assertion (A): Humans and chimpanzees share a common ancestor. [K] / [CO5]
Reason (R): DNA evidence shows ~98% similarity in sequences.

a). Both A and R are true, and R is the correct explanation of A

b). Both A and R are true, but R is not the correct explanation of A

c). A is true, but R is false

d). A is false, but R is true

18. What was a significant consequence for Galileo due to his strong support for Copernicus's heliocentric model? [U] / [CO1]

a). He received widespread immediate acceptance and accolades.

b). He was appointed as a chief advisor to the Church.

c). He faced conflict with the Church and was eventually tried for heresy.

d). He became the first person to travel to the Moon.

19. The uncertainty in measuring a pendulum's time period is reduced by [K] / [CO6]

a). Using a stopwatch with larger least count

b). Measuring the time for multiple oscillations

c). Taking a single measurement

d). Measuring at random times

20. Which statement about solar eclipses is a fact? [U] / [CO5]

a). Eclipses can cause natural disasters like earthquakes and tsunamis.

b). Animals behave strangely because an eclipse emits special rays that affect them

c). Looking at the Sun directly during an eclipse can cause permanent eye damage

d). Solar eclipses are rare and happen only once every few centuries.

21. Which of the following best describes a spurious measurement? [U] / [CO6]

a). A measurement taken with the smallest possible error.

b). A measurement that deviates significantly from the rest of the data set.

c). A measurement taken multiple times for consistency.

d). A measurement rounded off to the nearest integer

22. The first human in space was [K] / [CO2]

a). Yuri Gagarin b). Neil Armstrong

c). Valentina Tereshkova d). Buzz Aldrin

23. Which term refers to evidence supporting a theory? [U] / [CO1]

a). Corroboration

b). Contradiction

c). Hypothesis

d). Law

24. Graphene, which is a nanomaterial, is an allotrope of which element? [K] / [CO2]

a). Silicon b). Boron

c). Nitrogen d). Carbon

25. Which of the following is a negative environmental impact of science and technology? [U] / [CO3]

a). Electronic waste (E-waste) generation

b). Resource Depletion

c). Carbon Emissions from wide use of Technology

d). All of the above

26. The discovery that the universe is expanding shows that scientific theories can change when: [U] / [CO1]

a). Scientist makes random guesses

b). Ancient beliefs are followed strictly

c). Personal opinions of scientists change

d). New Evidence is found

27. Which early modern scientist's observations of the phases of Venus supported the heliocentric model, challenging the ancient geocentric worldview? [U] / [CO1]

a). Galileo Galilei

b). Tycho Brahe

c). Johannes Kepler

d). Nicolaus Copernicus

28. What is the main purpose of a nuclear power plant? [U] / [CO2]

a). To produce electricity

b). To produce coal

c). To produce natural gas

d). To produce solar energy

29. Assertion (A): Random errors shift all measured values in one direction. [U] / [CO6]
Reason (R): Averaging reduces random errors

a). Both A and R are true, and R is the correct explanation of A

b). Both A and R are true, but R is not the correct explanation of A

c). A is true, but R is false

d). A is false, but R is true

30. The term “scientific temper” was popularized in India by: [U] / [CO4]

a). Dr. A.P.J. Abdul Kalam b). Jawaharlal Nehru

c). Vikram Sarabhai d). C.V. Raman

[2x25 = 50]